

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Owned backend security of production AI platform: Auth0 RBAC, role-scoped API access, Sentry error monitoring, Terraform-provisioned infra; shipped 10+ features across 7 deployments in TypeScript. - Engineered multi-model LLM pipelines (Claude, GPT-4o, Gemini) via MCP; built document processing (PDF gen, email parsing, S3) and full Playwright test coverage.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Designed cloud-based genomic platform with JBrowse 2 on Google Cloud Run, enabling 100+ researchers to explore 500GB datasets worldwide. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Led model red-teaming and adversarial validation on production AI systems to identify failure modes and strengthen reliability. - Built 5+ AI-driven workflows including health claim automation with OCR and NLP; automated deployments on AWS and GCP.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built distributed Kafka + Spark + Hadoop pipelines for batch and real-time ingestion, cutting ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Demand Paging – Xinu OS	Systems Programming	Oct 2024 – Dec 2024
- Engineered 8-store virtual memory system in C for Xinu OS with Second-Chance page replacement; reduced page faults by 25%. - Added 7+ kernel system calls, increasing memory performance by 15% using inverted page tables and a custom fault handler.		
Glance AI	Privacy-First RAG	2025
- Built compliance-ready RAG system for healthcare with per-tenant vector isolation preventing cross-tenant data leakage and hallucinations. - Designed deterministic output architecture ensuring factual correctness for regulated, privacy-sensitive environments.		
Inventory Reservation Service	Backend API	2026
- REST API in Java 21/Spring Boot with PostgreSQL row-level locking; Spring Security RBAC, idempotent POST endpoints, RFC 9457 errors. - Concurrency verified with Testcontainers integration tests; Flyway migrations, multi-stage Docker build, GitHub Actions CI.		
LLM Eval Harness	Independent Project	2025
- Built vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini across 2,000 examples and 3 task types. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		

Languages and Technologies

- Languages:** C, Java, Python, TypeScript, JavaScript, SQL, Bash
- Systems & Security:** Kernel programming, virtual memory, RBAC, Auth0, red-teaming, TLS, OOP design patterns
- Backend & Cloud:** Spring Boot, Node.js, NestJS, PostgreSQL, Redis, AWS, GCP, Docker, Terraform, GitHub Actions
- AI & Tools:** Claude/GPT-4o/Gemini APIs, MCP, LangChain, OpenTelemetry, Prometheus, Testcontainers, Agile/Scrum

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.